

Normal Distribution

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Contour function

This function finds the contour function $\pi_x(\theta)$ given an observation x and the parameter of interest θ .

```
contour_normal <- function(x, theta){  
  x_bar <- mean(x)  
  n <- length(x)  
  return(2 * (pnorm(abs(x_bar-theta)*sqrt(n), lower.tail = F) ))  
}
```

Now we assume the true parameter is 0 and sample size is 1, let's plot the contour function given that we observed $X = 1$.

```
n <- 1  
true_theta <- 0  
# X <- rnorm(n) + true_theta  
X <- 1  
  
theta <- seq(true_theta-6, true_theta+6, .1)  
pi_normal <- contour_normal(X, theta)  
plot(theta, pi_normal, type = "l")  
pi_true_theta <- contour_normal(X, true_theta)  
abline(h=pi_true_theta, col = "red")
```

